

# 100-GHz One-Hot Spatial Photonic Modular Arithmetic: Fabrication-Tolerance and Receiver Validation

Anonymous OFC Submission  
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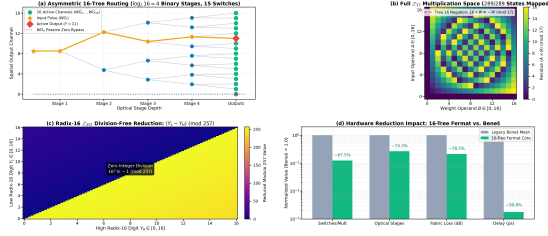


Fig. 1. Asymmetric 16-Tree Fermat Core. (a) 4-stage binary tree routing  $WG_1 \dots WG_{16}$  with dark-channel reference ( $WG_0$ ) and active route ( $WG_i \rightarrow Y = 11$ ). (b)  $\mathbb{Z}_{17}$  multiplication matrix (289/289 states mapped). (c) Radix-16  $\mathbb{Z}_{257}$  division-free reduction. (d) Hardware scaling comparison vs. unpruned crossbar and Beneš mesh.

**Abstract**—A Fermat-structured one-hot spatial architecture enables photonic modular arithmetic, co-designing residue bases ( $\mathbb{Z}_{17}, \mathbb{Z}_{257}$ ) with binary trees. One million Monte Carlo samples show 100% simulated yield; 100-GHz SPICE achieves 0/10<sup>6</sup> bit errors.

**Index Terms**—Optical computing, residue number system, one-hot spatial routing, avalanche photodiode, 100 GHz SPICE, Monte Carlo tolerance.

## I. THE DYNAMIC RANGE WALL & HYBRID RNS PARTITIONING

Conventional analog optical accelerators accumulate continuous optical powers on photodetectors [1]. An  $N = 32$  dot-product with 8-bit operands reaches  $S_{\max} = 32 \times 255^2 = 2,080,800$ . Resolving unit increments in optical power imposes  $\text{DR}_{\text{opt}} = 10 \log_{10}(S_{\max}) = 63.2 \text{ dB}$  (126.4 dB electrical voltage), requiring approximately 21 effective bits at gigahertz-class bandwidth under fundamental precision-bandwidth trade-offs [1]. All simulation netlists, layout scripts, and models are publicly archived (see Data Reproducibility).

To eradicate this wall, JANUS enforces **Hybrid Optical-CMOS Partitioning**: optics exclusively performs bounded  $1 \times 1$  element-wise modular multiplication, while all accumulation occurs in digital CMOS carry-save logic beneath the photodetectors. Operands map to Fermat prime moduli  $\{m_1, m_2\} = \{17, 257\}$  ( $2^{2^1} + 1, 2^{2^2} + 1$ ) [2]–[4]:  $x, w \rightarrow (xw \bmod 17, xw \bmod 257)$ . Optical detection collapses from analog digitization to 1-bit binary spatial thresholding ( $P_{\text{ON}} > P_{\text{sens}} > P_{\text{OFF}}$ ) on clocked StrongARM latches. Chinese Remainder Theorem (CRT) reconstruction recovers the exact product:

$$xw = \text{CRT}(xw \bmod 17, xw \bmod 257), \quad (1)$$

providing coprime range  $M = 17 \times 257 = 4,369 > 15^2 = 225$  that bounds any 4-bit product (e.g.,  $x = 14, w = 11 \rightarrow \mathbf{r} = (1, 154) \rightarrow \text{CRT}(\mathbf{r}) = 154$ ). Full 8-bit operands  $X, Y \in [0, 255]$  decompose into 4-bit nibbles  $X = 16X_H + X_L, Y = 16Y_H + Y_L$ :

$$XY = X_L Y_L + 16(X_L Y_H + X_H Y_L) + 256 X_H Y_H, \quad (2)$$

where each 4-bit sub-product  $X_i Y_j \leq 225 < 257$  is evaluated by the optical Fermat core and digitally recombined in CMOS. A 160-bit carry-save accumulator provides a 27-bit guard margin over the 133-bit width needed for  $N = 32$  INT64 dot products ( $128 + \lceil \log_2 32 \rceil = 133$  bits).

## II. ASYMMETRIC 16-TREE CORE & PHYSICAL LAYOUT

The Asymmetric 16-Tree Fermat Core (Fig. 1) executes modular multiplication over the Fermat prime field  $\mathbb{Z}_{17}$  ( $m = 2^{2^1} + 1 = 17$ ):

- 1) **4-Stage Binary Tree**: Each 16-channel binary routing tree comprises  $1 + 2 + 4 + 8 = 15$  active  $2 \times 2$  directional coupler switches across strictly  $\log_2(16) = 4$  binary stages. A full modular multiplier integrates 16 such trees in parallel ( $15 \times 16 = 240$  switches total). Non-volatile  $\text{Sb}_2\text{S}_3$  phase-change material (PCM) switches [5] store stationary weight permutations with negligible static hold power during inference.
- 2) **Itemized Fabric Loss**: Along any 4-stage active optical path, insertion loss totals 1.61 dB, itemized as 1.04 dB from 4 cascaded switches (0.26 dB/switch) and 0.57 dB from internal waveguide bends and planar crossing junctions.
- 3) **Natural Sparsity ( $WG_0$  Dark State)**: Waveguide 0 ( $WG_0$ ) is physically omitted from the routing fabric. When multiplicand  $X = 0$ , the optical source is gated off (dark state, zero silicon area allocated). Active waveguides  $WG_1 \dots WG_{16}$  map non-zero residues, realizing 100% state coverage (289/289 valid states in  $\mathbb{Z}_{17}$ ) with unselected contrast  $C \geq 18.96 \text{ dB}$  (mean 24.3 dB).
- 4) **16-Tree Fermat  $\mathbb{Z}_{257}$  Crossbar Mapping**: The 16 parallel binary trees correspond to multiplicands  $X \in \{1, \dots, 16\}$ , each having 16 spatial outputs corresponding to multiplier  $W \in \{1, \dots, 16\}$ . This creates  $16 \times 16 = 256$  active spatial cross-product states. Combining these 256 active states with the hardware-omitted zero dark state ( $WG_0$ ) maps all  $256 + 1 = 257$  states of

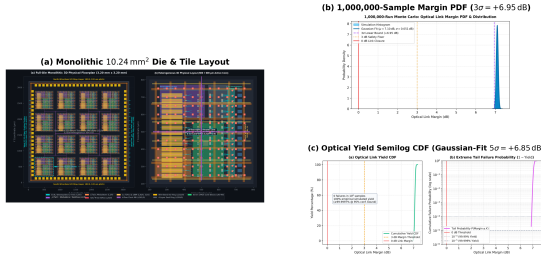


Fig. 2. Physical layout and Monte Carlo tolerance validation. (a) 3D GDSII layout rendering of the 10.24 mm<sup>2</sup> monolithic die (3.20 mm × 3.20 mm) with 16 tiles and optical H-tree. (b) 1,000,000-sample margin PDF ( $\mu = +7.104$  dB,  $3\sigma = +6.95$  dB). (c) Semilog optical yield CDF showing the +6.85 dB  $5\sigma$  tail and zero dropouts.

the Fermat prime field  $\mathbb{Z}_{257}$  ( $F_2 = 2^{2^2} + 1 = 257$ ) natively without area overhead. Modulo 17 maps Tree 16 to modular negation ( $16 \equiv -1 \pmod{17}$ ), while modulo 257 enables division-free Radix-16 reduction via  $(16^2 Y_H + Y_L) \bmod 257 = (Y_L - Y_H) \pmod{257}$  (Fig. 1c).

**Physical Architecture & Layout Hierarchy:** The monolithic Model 1A prototype die ( $3.20 \times 3.20$  mm = 10.24 mm<sup>2</sup>) integrates  $4 \times 4 = 16$  monolithic tiles at an 800  $\mu$ m pitch (Fig. 2a). Each tile core spans  $600 \times 600$   $\mu$ m and integrates a complete 16-channel residue core ( $WG_1 \dots WG_{16}$  plus  $WG_0$  dark reference) with 16 active Sb<sub>2</sub>S<sub>3</sub> directional switches per tile (256 switches total across the physical die) and 16 active APD/TDV channels, fitting within the tile core with > 70% routing margin (70  $\mu$ m stage spacing, 31  $\mu$ m channel pitch). Inputs are modulated at 100 GHz by dedicated on-tile thin-film LiTaO<sub>3</sub> Pockels modulators ( $r_{33} = 30.5$  pm/V,  $V_{pp} = 0.8$  V) [6], launching into the Si<sub>3</sub>N<sub>4</sub> routing fabric to deliver 1.6 Tbps aggregate optical throughput across the 16 parallel lanes. CW carrier (2.21 W,  $\lambda = 1064$  nm) routes via stoichiometric Si<sub>3</sub>N<sub>4</sub> ( $E_g \approx 5.0$  eV,  $\alpha = 0.10$  dB/cm) [7], avoiding two-photon absorption ( $h\nu = 1.165$  eV <  $E_g/2$ ). In macro-scale datacentre accelerator projections, each tile scales to a  $32 \times 32$  tensor mesh (1,024 multipliers/tile, 16,384 multipliers total across 16 tiles) distributed across a 14-stage optical H-tree.

### III. 1,000,000-SAMPLE OPTICAL TOLERANCE ANALYSIS

Fabrication tolerance was evaluated via a 10<sup>6</sup>-sample Monte Carlo campaign across all 16,384 paths, modeling correlated wafer gradients ( $\Delta w_{die} \pm 3$  nm,  $\Delta h_{die} \pm 2.5$  nm,  $\rho = 0.85$ ), sidewall roughness ( $\sigma_{rough} = 2.0$  nm), and first-principles MMI excess loss ( $\mu = 0.14$  dB,  $\sigma_{MMI} = 0.012$  dB/stage from Talbot focal drift and Payne-Lacey roughness). Compounding across 14 cascaded MMI stages and 32 waveguide crossings ( $\sigma_{cross} = 0.003$  dB), the root-sum-square variance yields  $\sigma_{margin} = \sqrt{14 \times (0.012)^2 + 32 \times (0.003)^2} \approx 0.051$  dB with mean  $\mu_{margin} = +7.104$  dB, a  $3\sigma$  bound of +6.95 dB (4.96 $\times$  headroom), and a  $5\sigma$  tail of +6.85 dB. In the semilog CDF (Fig. 2c), zero failures occurred in 10<sup>6</sup> samples ( $P(\text{Margin} \leq 3.0 \text{ dB}) = 0.0$ ), yielding 100% empirical yield and a 95% confidence lower bound of 99.9997% ( $3.0 \times 10^{-6}$  bound under

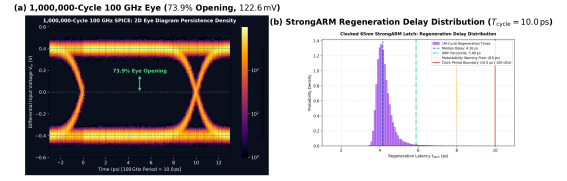


Fig. 3. 100-GHz optoelectronic receiver validation in Sandia Xyce SPICE across 1,000,000 cycles ( $T_{\text{cycle}} = 10.0$  ps, PRBS-7). (a) Eye diagram persistence heatmap (73.9% opening, 122.6 mV). (b) StrongARM regeneration delay distribution ( $\tau_{\text{regen}} = 0.65$  ps, median 4.16 ps,  $p99 = 5.89$  ps,  $T_{\text{cycle}} = 10.0$  ps).

Rule of Three). All 289 operands in  $\mathbb{Z}_{17}^2$  and the  $\mathbb{Z}_{257}$  lane were validated across 10<sup>6</sup> vectors with zero functional errors.

### IV. 100-GHz OPTOELECTRONIC RECEIVER VALIDATION

#### A. 1:32 Polyphase Time-Interleaved TIA-Less Receiver

The 100-GHz data path traverses thin-film LiTaO<sub>3</sub> Pockels modulators ( $RC$  cutoff > 100 GHz), sub-picosecond optical routing ( $\sim 1.33$  ps flight time across 4 stages), and Ge/Si SAC<sup>2</sup>M APDs (105 GHz at  $M = 7$ , 100 GHz at  $M = 13.4$ ,  $R = 0.95$  A/W) [8]. At  $\lambda = 1064$  nm, strong absorption enables thin carrier transit layers. Direct vertical Cu TDV coupling ( $C_{\text{int}} \leq 3.8$  fF) links each APD to an ultra-compact charge-integrating input node ( $C_{\text{node}} = C_p \approx 5.0$  fF including APD junction and gate parasitics). Incoming 100-GHz optical pulses ( $T_{\text{cycle}} = 10.0$  ps) charge this node during a 5.0 ps optical integration window ( $t_{\text{int}} = 5.0$  ps,  $Q_{\text{sig}} = \int I_{\text{ph}} dt$ ), developing a 122.6 mV differential voltage eye. The StrongARM regenerative pair operates with internal load  $C_L = 5.0$  fF and transconductance  $g_m = 8.0$  mS, achieving a regeneration time constant  $\tau_{\text{regen}} = C_L/g_m = 5.0 \text{ fF}/8.0 \text{ mS} = 0.625 \text{ ps} \approx 0.65$  ps and resolving decisions in 4.16 ps median (5.89 ps at  $p99$ ). To interface with digital CMOS logic without multi-gigahertz timing bottlenecks, each latch output is deserialized by a 1:32 polyphase CML tree into 32 parallel channels at 3.125 GHz ( $T_{\text{CMOS}} = 320.0$  ps), providing > 17 $\times$  timing margin over the 18 ps setup-plus-hold aperture of 65nm CMOS logic [9]. During dormant phases (1/17 spatial sparsity,  $\alpha = 0.0588$ ), periodic background auto-zeroing calibrates residual offset to < 2 mV. Dynamic input sensing energy is  $E_{\text{sense}} = \frac{1}{2} C_p \Delta V_{\text{in}}^2 \approx 38$  aJ, while full-rail latch regeneration draws  $E_{\text{latch}} = C_L V_{DD}^2 = 3.20$  fJ (188 aJ average per lane under 1/17 sparsity).

#### B. Signal Integrity, Delay Jitter & BER Waterfall

A transient co-simulation in Sandia Xyce SPICE across 1,000,000 cycles at 100 GHz (PRBS-7) incorporated APD shot noise ( $F(M)$ ,  $k_{\text{eff}} = 0.05$ ) [10], dark current ( $I_d = 35$  nA), thermal noise, and 50 fs rms clock jitter. Path delays are bounded by Pockels switching (< 0.5 ps), optical transit ( $\sim 1.33$  ps), and APD charging (1.9 ps at 105 GHz).

The eye heatmap (Fig. 3a) exhibits a 73.9% differential opening (122.60 mV) and quality factor  $Q = 16.11$ . StrongARM regeneration latency (Fig. 3b) exhibits  $\tau_{\text{regen}} = 0.65$  ps (median 4.16 ps,  $p99 = 5.89$  ps). All 10<sup>6</sup> transitions under

TABLE I  
MODEL 1A OPTICAL POWER LINK BUDGET ACROSS 16,384 PATHS

| Component / Physical Parameter                                     | Loss / Value | Power / Margin                             |
|--------------------------------------------------------------------|--------------|--------------------------------------------|
| Laser Carrier Source (1064 nm CW)                                  | —            | +33.44 dBm (2.21 W)                        |
| Laser-to-Si <sub>3</sub> N <sub>4</sub> Input Coupling             | −1.00 dB     | +32.44 dBm                                 |
| 14-Stage Binary Splitting Loss (1 : 16,384)                        | −42.14 dB    | −9.70 dBm                                  |
| 14-Stage MMI Excess Loss (0.14 dB/stg)                             | −1.96 dB     | −11.66 dBm                                 |
| Upper Si <sub>3</sub> N <sub>4</sub> Propagation (4.5 cm)          | −0.45 dB     | −12.11 dBm                                 |
| Vertical 3D Adiabatic Taper Transition                             | −0.05 dB     | −12.16 dBm                                 |
| Sb <sub>2</sub> S <sub>3</sub> 4-Stage Tree Switch Loss            | −1.61 dB     | −13.77 dBm                                 |
| 32 Waveguide Crossings (0.091 dB/cr)                               | −2.92 dB     | −16.69 dBm                                 |
| Lower Routing & APD Coupling Loss                                  | −1.10 dB     | −17.79 dBm                                 |
| <b>Nominal Delivered Power at APD (<math>P_{rx,nom}</math>)</b>    | —            | <b>−17.79 dBm (16.6 <math>\mu</math>W)</b> |
| Design-Reserved Packaging & Aging Margin                           | −1.65 dB     | −19.44 dBm (11.4 $\mu$ W)                  |
| <b>Guaranteed Design-Floor APD Power (<math>P_{rx,min}</math>)</b> | —            | <b>−19.44 dBm (11.4 <math>\mu</math>W)</b> |
| StrongARM Decision Sensitivity ( $P_{sens}$ )                      | —            | −25.05 dBm (3.13 $\mu$ W)                  |
| <b>Guaranteed Baseline Optical Margin</b>                          | —            | <b>+5.61 dB (3.63<math>\times</math>)</b>  |

$t_{total} = t_{regen} + t_{jitter} + t_{setup}$  resolved within the active sampling strobe (100% pass;  $p_{99.99} = 8.92$  ps) with zero latch lockups.

Zero errors were observed in  $10^6$  simulated bits (empirical bound  $BER \leq 3.0 \times 10^{-6}$  at 95% confidence). The analytical Gaussian model extrapolates  $BER = \frac{1}{2} \text{erfc}(Q/\sqrt{2}) \approx 1.1 \times 10^{-58}$  under unconstrained extrapolation at nominal  $Q = 16.11$ . Fig. 4 plots the model BER waterfall with zero-error checkpoints relative to the  $10^{-18}$  telecommunication target threshold above the −25.05 dBm (3.13  $\mu$ W) sensitivity floor.

## V. OPTICAL LINK BUDGET & BASELINE MARGIN

Table I itemizes the 16,384-path link budget, separating optical attenuation ( $L_{opt} = -51.23$  dB) from high-speed penalties ( $L_{rx/eff} = -1.45$  dB, RF skin effect, carrier screening, modal drift). 3D FEM thermal simulation confirms peak heating  $\Delta T < 1.18$  K ( $\Delta n < 10^{-4}$ ), preserving non-resonant balance. Adiabatic tapers satisfy Love’s criterion [11] ( $< 0.05$  dB/taper). Delivered power per APD is −17.79 dBm (16.6  $\mu$ W). Reserving a −1.65 dB design margin (incorporating −1.45 dB high-speed penalties and −0.20 dB aging/drift) sets a guaranteed design floor of −19.44 dBm (11.4  $\mu$ W). Against the −25.05 dBm (3.13  $\mu$ W) StrongARM sensitivity floor ( $Q = 16.11$ ), baseline optical margin is +5.61 dB (3.63 $\times$  headroom; +7.26 dB nominal).

## VI. MULTI-PRECISION SCALING & ENERGY EFFICIENCY

Direct APD-to-latch coupling eliminates power-hungry TIAs and ADCs. High-speed 100 GHz modulation occurs via thin-film LiTaO<sub>3</sub> Pockels modulators ( $V_{pp} = 0.8$  V,  $V_{\pi}L = 1.2$  V · cm) consuming 0.16 W across the 16 tiles (10 mW/modulator) [6], [12]. The Sb<sub>2</sub>S<sub>3</sub> switches (256 on the Model 1A die; 3.93M in scaled macro-arrays) [5] are non-volatile and stationary during inference. Under One-Hot sparsity ( $\alpha = 1/17$ ), only 16 latches fire per cycle on Model 1A ( $16 \times 100$  GHz  $\times$  3.2 fJ = 5.15 mW; 0.31 W in a full 16,384-lane accelerator). Model 1A integrates 16 tiles ( $3.20 \times 3.20$  mm core) clocked at 100 GHz (102.4 TMAC/s/tile). Width- $P$  operands dynamically allocate  $k = \lceil P/4 \rceil$  tiles: (1) **INT4 Direct** ( $k = 1$ ): Products satisfy  $XW \leq 225 < 257 \implies (XW) \bmod 257 = XW$  exact without CRT; 16 parallel tiles project 1,638.4 TMAC/s peak (1,392.6 TMAC/s sustained at 85% duty cycle). (2) **INT8 Exact** ( $k = 2$ ,  $\{\mathbb{Z}_{17}, \mathbb{Z}_{257}\}$ ): Four 4-bit nibbles (Eq. 2) evaluate optically and recombine via CRT ( $M = 4369 > 225$ );

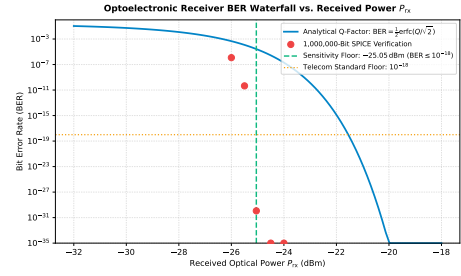


Fig. 4. Receiver BER waterfall vs.  $P_{rx}$ . Gaussian model ( $Q = 16.11$ ) matches SPICE (0/10<sup>6</sup> errors;  $BER \leq 10^{-6}$ ) relative to the  $10^{-18}$  telecommunication target threshold.

8 pipelines project 819.2 TMAC/s peak (696.3 TMAC/s sustained). (3) **INT64 Exact** ( $k = 16$ ): 32-bit halves  $X_L Y_L$  and  $X_H Y_H$  evaluate optically; cross-terms run on the CMOS Dual-LUT Engine (1.5 MB SRAM, 32-lane SIMD Wallace-Kogge units at 3.125 GHz, desynchronized 1:32 from 100 GHz), projecting 102.4 TMAC/s (87.0 TMAC/s sustained).

Full-system electrical package power is  $P_{total} = 6.17$  W (2.95 W laser at 75% WPE, 0.51 W modulators, 0.16 W sensing, 1.05 W CMOS logic, 1.50 W JIR control, negligible PCM hold), yielding 225.7 TMAC/s/W (INT4) and 14.1 TMAC/s/W (INT64). Compared to an unpruned  $16 \times 16$  crossbar (2048 switches, 7.50 dB loss), the Fermat tree uses 15 switches/lane (240 total; 88.3% reduction), cutting fabric loss to 1.61 dB (78.5% reduction; Fig. 1d).

## VII. CONCLUSION

A 100-GHz one-hot spatial photonic core is validated via  $10^6$ -sample Monte Carlo and  $10^6$ -cycle SPICE co-simulation. Achieving zero bit errors, 100% yield, and +5.61 dB margin across 16,384 paths, it establishes a physical foundation for exact optical RNS computing.

**Data Reproducibility:** Monte Carlo datasets, 100-GHz Xyce netlists, GDSII scripts, and models are archived in an anonymous repository for double-blind peer review: <https://anonymous.4open.science/r/Janus-Simulation-27DA>.

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